# Girls Build Bots — Brownie Robotics Badge Curriculum

## A badge-aligned workshop guide for robotics outreach events

**Level:** Brownies (Grades 2–3) · **Badges covered:** Programming Robots, Designing Robots, Showcasing Robots **Format:** One full-day camp (5 hours) or three 90-minute sessions · **Ratio:** 1 mentor per 3–4 Scouts

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Robotics badge series. Official badge requirements are available in your Girl Scout Volunteer Toolkit (VTK). Please verify completed activities against the official steps before awarding badges. Badges are purchased by troops through your council shop.

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## Big Ideas for Brownies

At this age, robotics is about **discovery and play**. Brownies should leave believing three things:

1. Robots are machines that **sense, think, and act** — and people (like them!) tell robots what to do.
2. A program is just **a list of steps in order** — and they can write one.
3. Engineers **try, fail, and try again** — mistakes are part of building.

**Vocabulary to introduce:** robot, program, sensor, command, sequence, engineer, prototype.

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## Module 1: Programming Robots (90 minutes)

**Badge theme:** Understanding what a program is and creating one a robot could run.

### Materials

- Painter's tape (floor grids)
- Arrow cards (forward, left turn, right turn, stop) — printable, ~30 sets
- Laptops or tablets with **Microsoft MakeCode Arcade** (arcade.makecode.com — free, browser-based)
- Optional: any simple floor robot (Bee-Bot, Code & Go Mouse, Sphero Indi) if the team owns them

### Activity 1.1 — "Robot Says" Warm-Up (15 min)

Mentors act as "robots" that only follow *exact* commands. Scouts give instructions to walk a mentor across the room ("take 3 steps forward"). When Scouts say something vague ("go over there"), the robot-mentor freezes with an error beep. **Debrief:** robots need clear, step-by-step instructions — that's a program.

### Activity 1.2 — Human Grid Maze (25 min)

Tape a 5×5 grid on the floor with a start, a goal (a stuffed animal to "rescue"), and 2–3 obstacles. In pairs, one Scout is the robot, one is the programmer. The programmer arranges arrow cards into a sequence, *then* the robot executes it exactly. If the robot hits an obstacle: debug the card sequence and re-run.

**Key teaching moment:** name the loop — *plan → run → find the bug → fix it → run again.*

### Activity 1.3 — First MakeCode Arcade Program (35 min)

Scouts follow a mentor-led build of a simple "sprite walk" program: make a character, move it with arrow keys, collect a star, play a sound. Mentors demonstrate one block at a time on a projector; Scouts mirror on their own devices. Fast finishers: change the character, add a second star, change the sound.

### Wrap-Up Circle (15 min)

Each Scout shows one thing her program does. Ask: "What did you fix when it didn't work the first time?"

### Completion Checklist — Module 1

- ☐ Can explain that a program is a list of steps a robot follows
- ☐ Gave clear step-by-step commands to a "human robot"
- ☐ Built and debugged an arrow-card sequence through the maze
- ☐ Created and ran a simple program in MakeCode Arcade

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## Module 2: Designing Robots (90 minutes)

**Badge theme:** Learning what robots do for people and designing a robot that solves an everyday problem.

### Materials

- Craft supplies: cardboard, paper cups, pipe cleaners, googly eyes, tape, foil, straws, bottle caps
- "Robot jobs" picture cards (vacuum robot, Mars rover, factory arm, hospital delivery robot, robot dog)
- Design sheets: "My Robot's Name / My Robot's Job / What It Senses / What It Does"

### Activity 2.1 — What Do Robots Do? (15 min)

Sort the robot job cards: *robots that help at home / robots that explore / robots that work in factories / robots that keep people safe.* Ask each Scout: "What job do YOU wish a robot could do for you?"

### Activity 2.2 — Sense–Think–Act Game (15 min)

Play a movement game: mentor calls a "sensor input" ("you see a wall!" / "you hear a clap!" / "it's dark!") and Scouts act out the robot response they choose ("turn around," "stop," "turn on lights"). Connects sensing to action in their bodies before they build.

### Activity 2.3 — Build a Prototype Bot (45 min)

Each Scout picks an everyday problem (messy room, walking the dog, watering plants, carrying a backpack) and builds a **craft prototype** of a robot that solves it. Mentors circulate asking the three engineer questions: *What does it sense? What does it think about? What does it do?* Scouts complete their design sheet.

**Mentor note:** the prototype does not need to function — this badge level is about design thinking, not mechanisms. Praise ideas and iteration, not neatness.

### Wrap-Up (15 min)

Gallery walk: prototypes on the table, Scouts walk around with one sticky-note compliment to leave on someone else's robot.

### Completion Checklist — Module 2

- ☐ Can name jobs robots do for people
- ☐ Can explain "sense, think, act" in her own words
- ☐ Designed a robot that solves an everyday problem
- ☐ Built a prototype and completed a design sheet

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## Module 3: Showcasing Robots (90 minutes)

**Badge theme:** Sharing your robot design with others and learning how robots are shared with the world.

### Materials

- Prototypes and design sheets from Module 2
- "Showcase station" table with tablecloth and name cards
- Optional: the FRC team's competition robot for a demo (the showstopper!)

### Activity 3.1 — Meet a Real Robot (20 min)

The high school team demos their FRC robot: what it senses, what its program tells it to do, what job it does in competition. Scouts ask questions. **This is often the most-remembered moment of the whole camp — budget time for it.**

### Activity 3.2 — Prepare Your Showcase (30 min)

Each Scout prepares a 60-second "robot reveal" using a simple frame taped to the table: 1. "My robot's name is…" 2. "The problem it solves is…" 3. "It senses… and then it…" 4. "If I built it again, I would change…"

Mentors run practice rounds in pairs. Coach volume, eye contact, and holding the prototype up.

### Activity 3.3 — The Showcase (30 min)

Scouts present to the group (and to parents/troop leaders at pickup if scheduled). Every presenter gets a round of applause. Mentors ask one friendly follow-up question each.

### Wrap-Up (10 min)

Badge ceremony moment: troop leaders confirm completion against the VTK and announce that Scouts have finished the badge work.

### Completion Checklist — Module 3

- ☐ Saw a working robot and asked a question about it
- ☐ Prepared a short presentation about her own robot design
- ☐ Presented to an audience
- ☐ Reflected on one thing she would improve

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## Mentor Guide (High School Robotics Students)

- **Never touch the Scout's project.** Hands behind your back; guide with questions ("What happens if you try…?").
- **Use wait time.** Count to five after asking a question. Brownies will get there.
- **Celebrate bugs.** Say "you found a bug — that's what real programmers do all day" every single time.
- **Watch the quiet ones.** Rotate to Scouts who aren't asking for help; they need you most.
- **Energy management:** Brownies fade after ~25 minutes of one activity. Move to the next segment on schedule even if unfinished.

## Pre/Post Survey (Impact Measurement)

Ask each Scout before and after camp (thumbs up / sideways / down): 1. I know what a robot program is. 2. I could design a robot to solve a problem. 3. I want to do more robotics. 4. I could teach a friend something about robots.

Record counts; report the before→after change as your outcome metric.

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*Created for the “Girls Build Bots” Girl Scout Gold Award project. This is an original curriculum aligned to the Brownie Robotics badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*